Background Information:

Exploratory subgroup analyses might be conducted in phase 2 or phase 3. The goal of those subgroup analyses on clinical trial data might cover one or multiple goals including assessing treatment effect heterogeneity, performing (predictive) variable selection, assigning individuals to subgroup(s), estimating subgroup proportion or estimating treatment effects in subgroup(s).

The underlying assumption is that some patient (groups) profit more from the experimental drug in comparison to control/standard of care (SoC) treatment than others. “Profit more” could refer to qualitative interactions with treatment (i.e. no benefit or even harm when receiving experimental instead of SoC for those not belonging to subgroup) as well as quantitative interactions (i.e. all benefit from receiving experimental instead of SoC but the treatment effect is higher in a subgroup). From a different perspective a related task is to define groups of patients with treatment effect homogeneity within subgroups but treatment effect heterogeneity between subgroups. To simplify the setting, we aim here to declare one “profiting” subgroup, and the rest form the “non-profiting” complement.

Simulated data sets are set up to resemble randomized clinical trial data sets. Each data set should be considered as representing an independent trial. Except for the standardized format (see below) there is no connection between the data sets. As the primary objective of those trials is usually identifying a common effect on the primary endpoint in the trial population, we performed sample size calculations accordingly. Binary and continuous endpoints are considered. For both endpoints, greater values refer to more favorable outcomes/ response. For the binary endpoint, it means 1 refers to treatment responders and 0 to non-responders. Each data set will only contain one endpoint.

The minimum relevant effect for a difference in rates (binary endpoint) for the target population in phase 3 is 0.1.
The minimum relevant effect for a difference in means (continuous endpoint) for the target population in phase 3 is 0.9 (individual standard deviation: 4.5).

Sample size planning with R package rpact is attached. Note, the critical value on treatment effect scale from phase 2 is (roughly) the expected effect for phase 3. The critical value on treatment effect scale from phase 3 is slightly below the minimum relevant effect for the full population.

Although the sample size planning for phase 2 and 3 trials is connected all simulated data sets are to be considered as independent as described above. The phase 2 and 3 setting is just chosen to derive a realistic justification for a sample size that is based on knowledge that is also available in practice.

All data sets contain one row per patient. Thus, the number of rows is equal to the number of patients +1 (for the variable names).

Number of Subjects

Binary endpoint: 150 and 500 (corresponding to phase 2 and phase 3)
Continuous endpoint: 150 and 700 (corresponding to phase 2 and phase 3).

Number of Covariables

We consider 6, 15 or 50 baseline covariables. They can be continuous or binary. A covariable could represent clinical characteristics or biomarker of any kind. We assume that a certain pre-selection of biomarkers was already done beforehand. Overall, there is no priority and no stronger/weaker biological rational for one or the other variable. All have equal prior probabilities of being prognostic or predictive.

Allocation Ratio

Treatment allocation is fixed (i.e. perfect randomization) with a 1:1 ratio. Randomization was not stratified.

Missing Values

There are no missing values in the datasets.

Naming Conventions

The response is represented as “Y”, treatment as “W” (1=experimental, 0=control), and covariables should appear as “X{i}”. The patient identifier is “ptid” is 1 up to the sample size for the data set. When returning data sets please have them sorted according to ptid.

Process

Input (you receive from us)	**Output (we receive from you)**
A **sheet with additional guidance** and explanations beyond information provided on the web page (this document)	
A **sheet to indicate which of the tasks** are addressed*	Return **sheet indicating which of the tasks** you addressed*
A zip-file with **30 data sets** in csv format	Return a ZIP file containing the **30 data sets** with patient-level subgroup assignments **added as an extra column**, using a variable 'S' where 1 indicates subgroup and 0 indicates complement
An empty **results file** that allows to collect dataset-level information	Return **results file** with the collected dataset-level information if corresponding tasks are selected.**
	Provide a **short summary** of the methods/algorithms applied for each task that was addressed. If multiple approaches were used for a task, **indicate how the final decision was derived** (e.g. pure algorithm based or agreement among team members)

* Please use a browser like Chrome or Edge if Adobe does not correctly display the input mask

** If you don’t use MS Excel to edit the “Submission_Results.xlsx” file, you can for instance use the read_excel function from the R package readxl and send us the final results table in a suitable format (e.g., rds or Rdata).

Evaluation

- **Task 1: "Assessing treatment effect heterogeneity"**
  Agreement rate over all data sets
- **Task 2: "Performing variable selection"**
  Average score with score per data set based on proportion of correctly classified variables into neither/predictive/prognostic/both plus a bonus of 0.25 if the exact correct set of predictive variables is selected. Note, declaring predictive as both or vice versa is considered as correct decision for the bonus.
- **Task 3: "Assigning individuals"**
  Average over all agreement rates per data set
- **Task 4: "Estimating subgroup proportion"**
  Average difference between estimated proportion and true rate per data set
- **Task 5: "Estimating treatment effect"**
  Average root mean square error (RMSE)
- Task specific scores will be combined into a weighted Score $\Rightarrow$ Winner
  - To be included in the rating for each task, at least 80% of the provided data sets must have been analysed

Selection of teams for presenting at ISCB/GMDS conference as well as for a publication might not only depend on the ranking of the overall score but also on

- Task-specific ranking
- Methods applied
- Decision making process
- Whether simple subgroup rules were provided

Sharing Data Generating Model

The data generating model will be presented at ISCB/GMDS conference together with the evaluation of the results. The corresponding R code will be shared afterwards with the submitting participants.

Appendix

Trial planning for binary and continuous endpoint

```
pl2 <- getSampleSizeRates(getDesignGroupSequential(kMax = 2,typeOfDesign =
"noEarlyEfficacy", informationRates=c(0.3,1),futilityBounds=0.15,alpha = 0.
025, sided=1,beta=0.2), groups = 2, pi1 = c(0.63),
                    pi2 = 0.5, allocationRatioPlanned = 1)


pl4 <- getSampleSizeRates(getDesignGroupSequential(kMax = 1, alpha = 0.05,
sided=1,beta=0.2), groups = 2, pi1 = c(0.7),
                    pi2 = 0.5, allocationRatioPlanned = 1)




knit_print(summary(pl2))
```

Sample size calculation for a binary endpoint

Sequential analysis with a maximum of 2 looks (group sequential design), one-sided overall significance level 2.5%, power 80%. The results were calculated for a two-sample test for rates (normal approximation), H0: pi(1) – pi(2) = 0, H1: pi(1) = 0.63, control rate pi(2) = 0.5.

Stage	1	2
Planned information rate	30%	100%
Cumulative alpha spent	0	0.0250
Stage levels (one-sided)	0	0.0250
Efficacy boundary (z-value scale)	Inf	1.960
Futility boundary (z-value scale)	0.150	
Efficacy boundary (t)		0.088
Futility boundary (t)	0.012	
Cumulative power	0	0.8000
Number of subjects	149.1	**497.0**
Expected number of subjects under H1		471.7
Overall exit probability (under H0)	0.5596	
Overall exit probability (under H1)	0.0728	
Exit probability for efficacy (under H0)	0	
Exit probability for efficacy (under H1)	0	
Exit probability for futility (under H0)	0.5596	
Exit probability for futility (under H1)	0.0728	

Legend:

- *(t)*: treatment effect scale

```
knit_print(summary(pl4))
```

Sample size calculation for a binary endpoint

Fixed sample analysis, one-sided significance level 5%, power 80%. The results were calculated for a two-sample test for rates (normal approximation), H0: pi(1) – pi(2) = 0, H1: pi(1) = 0.7, control rate pi(2) = 0.5.

Stage	Fixed
Stage level (one-sided)	0.0500
Efficacy boundary (z-value scale)	1.645
Efficacy boundary (t)	0.135
Number of subjects	**146.3**

Legend:

- *(t)*: treatment effect scale

```
plnv1 <- getSampleSizeMeans(getDesignGroupSequential(kMax = 2,typeOfDesign
= "noEarlyEfficacy", informationRates=c(0.3,1),futilityBounds=0.15, alpha =
0.025, sided=1,beta=0.2), groups = 2, alternative=1,
                            stDev = 4.5, allocationRatioPlanned = 1)
knit_print(summary(plnv1))
```

Sample size calculation for a continuous endpoint

Sequential analysis with a maximum of 2 looks (group sequential design), one-sided overall significance level 2.5%, power 80%. The results were calculated for a two-sample t-test, H0: mu(1) – mu(2) = 0, H1: effect = 1, standard deviation = 4.5.

Stage	**1**	**2**
Planned information rate	30%	100%
Cumulative alpha spent	0	0.0250
Stage levels (one-sided)	0	0.0250
Efficacy boundary (z-value scale)	Inf	1.960
Futility boundary (z-value scale)	0.150	
Efficacy boundary (t)		0.669
Futility boundary (t)	0.093	
Cumulative power	0	0.8000
Number of subjects	209.3	**697.8**
Expected number of subjects under H1		662.2

Stage	1	2
Overall exit probability (under H0)	0.5596	
Overall exit probability (under H1)	0.0728	
Exit probability for efficacy (under H0)	0	
Exit probability for efficacy (under H1)	0	
Exit probability for futility (under H0)	0.5596	
Exit probability for futility (under H1)	0.0728	

Legend:

- *(t)*: treatment effect scale

```
plnv3 <- getSampleSizeMeans(getDesignGroupSequential(kMax = 1, alpha = 0.1,
sided=1,beta=0.3), groups = 2, alternative=1.35,
                            stDev = 4.5, allocationRatioPlanned = 1)

knit_print(summary(plnv3))
```

Sample size calculation for a continuous endpoint

Fixed sample analysis, one-sided significance level 10%, power 70%. The results were calculated for a two-sample t-test, H0: mu(1) - mu(2) = 0, H1: effect = 1.35, standard deviation = 4.5.

Stage	Fixed
Stage level (one-sided)	0.1000
Efficacy boundary (z-value scale)	1.282
Efficacy boundary (t)	0.960
Number of subjects	**145.8**

Legend:

- *(t)*: treatment effect scale